Product Review Sentiment Analysis Using Machine Learning

Project Submission Report

Domain: Electrical / Electronics / Computer Products

1. Abstract

This project demonstrates a machine-learning pipeline that classifies product review text into positive, negative and neutral sentiment. Text is transformed using TF-IDF and classified with Logistic Regression. A Streamlit interface supports single-review prediction and dataset exploration.

2. Problem Statement

Product platforms receive large volumes of textual feedback. Manual review is slow and inconsistent. Automated sentiment classification can help summarize customer opinion.

3. Objectives

Prepare review text; generate TF-IDF features; train a three-class classifier; evaluate using a held-out split; provide an interactive dashboard.

4. Dataset and ethical disclosure

The bundled 300-row CSV is synthetic demonstration data created for immediate reproducibility. It is not collected from Amazon, Flipkart, Croma, Reliance Digital or Best Buy. Replace it with a genuine dataset that permits academic use before presenting findings as customer-review analysis.

5. Methodology

The data is split into training and test subsets using stratification and a fixed random seed. TF-IDF uses unigrams and bigrams. Logistic Regression is trained with balanced class weights. Evaluation includes accuracy, precision, recall, F1-score and a confusion matrix.

6. Architecture

CSV input → preprocessing → TF-IDF vectorization → Logistic Regression → sentiment prediction → Streamlit dashboard. The architecture diagram is included in docs/architecture.png.

7. Implementation

src/train.py trains and evaluates the model and writes the serialized pipeline and output metrics. app.py loads the saved pipeline, accepts a review and displays the predicted sentiment and class probabilities.

8. Results

On the bundled synthetic demonstration dataset, accuracy is 1.000 with 240 training rows and 60 held-out test rows. These results validate execution only and must not be interpreted as expected performance on real customer reviews.

9. Limitations

Synthetic language is simpler and more repetitive than genuine customer feedback. The model may not handle sarcasm, mixed sentiment, spelling errors, domain shift or multilingual text reliably.

10. Future Enhancements

Use a larger authentic licensed dataset; perform cross-validation and error analysis; add product-level comparisons; explore transformer-based models and aspect-based sentiment.

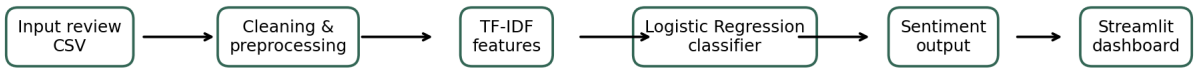
11. Reproduction

Install requirements with pip install -r requirements.txt; run python src/train.py; then run streamlit run app.py.

12. Submission links

GitHub repository: [Add repository URL]
Demo application: [Add deployed app URL]
Demo video: [Add video URL]
Collaborator: gundimedavenugopal (invite through GitHub repository settings).

Appendix: System Architecture



Training / evaluation: stratified 80:20 split • accuracy, precision, recall, F1-score, confusion matrix